

THEMES

Risk

PRINCE2 definition for Risk is as follows

Risk is a set of events that, should they occur, will have an effect on achieving the project objectives.

Another definition of Risk is:

Risk is an uncertain event that if it occurs, will have a positive or negative effect on a project objective.

Risk

The purpose of the Risk theme is to identify, assess and control uncertainty and, as a result, improve the ability of the project to succeed.

A Risk is an uncertain event that can impact the outcome of the project negatively or positively

Effective management of risks prevents them from turning into issues and this is the primary aim of risk theme. The risk management involves the process of identifying, assessing planning and implementing the risk response plan.

A key responsibility of a project manager is to continuously anticipate risk and respond to them so that they are not converted into issues.

The risks can be positive as well as negative. The positive risks are based on opportunities and strengths; however the negative risks are based on weakness and threats.

Risk

How to Express the Risk?

The risk is always expressed in terms of three variables.

- **Cause - What is the original cause of the Risk ?**
- **Event - What is the threat ?**
- **Effect - What is the risk ?**

Farmers' crops might get damaged due to heavy rain, as fields will get flooded.

What is original cause? The cause is heavy rain.

What is the threat? The threat is that fields might get flooded

What is the risk? The effect if the risk does happen is that the crops will get damaged.

Risk Management Procedure

The PRINCE2 procedure for risk management is summarized as follows;



Risk

Identify. First the context of the project is determined to understand the specific objectives that are at risk, and develop the risk management strategy.

Assess. The qualitative risk analysis is done by determining the **probability** (Likely hood of happening), **impact and Proximity** (How soon a Risk might materialize) of the risks. Based on the multiplication of these two factors urgency of the risk is decided and risk profiling is done

After the risk profiling is done. The quantitative risk analysis is conducted for high priority risks only as it is an expensive process. Techniques such as expected monetary value analysis, decision tree and sensitivity analysis using tornado diagrams can be used for quantitative risk analysis.

Risk

Probability and Impact Matrix

		Threats					Opportunities				
*Impact Probability		Very Low	Low	Medium	High	Very High	Very High	High	Medium	Low	Very Low
		0.05	0.10	0.20	0.40	0.80	0.80	0.40	0.20	0.10	0.05
Very High 71-90%	0.90	0.05	0.09	0.18	0.36	0.72	0.72	0.36	0.18	0.09	0.05
High 51-70%	0.70	0.04	0.07	0.14	0.28	0.56	0.56	0.28	0.14	0.07	0.04
Medium 31-50%	0.50	0.03	0.05	0.10	0.20	0.40	0.40	0.20	0.10	0.05	0.03
Low 11-30%	0.30	0.02	0.03	0.06	0.12	0.24	0.24	0.12	0.06	0.03	0.02
Very Low up to 10%	0.10	0.01	0.01	0.02	0.04	0.08	0.08	0.04	0.02	0.01	0.01

Risk

Qualitative Risk Analysis Matrix

Likelihood	Consequences				
	Insignificant	Minor	Moderate	Major	Severe
Almost certain	M	H	H	E	E
Likely	M	M	H	H	E
Possible	L	M	M	H	E
Unlikely	L	M	M	M	H
Rare	L	L	M	M	H

Risk

Plan

This is where appropriate responses for each of the threats and opportunities are identified in order to reduce the former and maximize the latter. Following are the strategies for negative and the positive risks.

Negative Risks

Avoid

Reduce

Transfer

Accept

Fallback

Share

Positive Risks

Enhance

Exploit

Reject

Share

Risk

Plan – Negative Risks

Avoid. This entails taking some action upfront and hence changing some aspect of the project such that the risk probability becomes zero and/or there will be no impact.

Reduce. Another term for this is mitigating the risk and unlike avoid, taking action to reduce will either reduce probability of happening or impact of the risk.

Transfer. The risk is transferred to a third party by making it or responsible for all or some of the financial impact of the risk, and this is normally done in the form of contract clauses that come into force as a result of such a risk.

Share . This response is a form of risk sharing between two or more parties and is normally built into a contract.

Risk

Plan – Negative Risks

Accept. This means taking no response action. This is usually the response strategy if severity of the risk is less than the cost or complexity of implementing a response action.

Fallback. This is also called as contingency planning and is different to the first three in that no action is taken up front. This is a reactive approach and entails creating a fallback plan with actions to be implemented only if the risk occurs. For Example a business continuity and disaster recovery plan.

Risk

Plan – Positive Risks

Exploit. This entails taking some action upfront that will seize the opportunity ensuring that it will occur and that the positive impact will be realized.

Enhance. Enhancing a risk involves identifying the root cause of a positive risk so that you can influence the root cause to increase the probability of happening of the positive risk.

Reject. It also means you are acknowledging that you'd rather not Exploit, Share, or Enhance the risk. This is normally chosen much like the accept response to threats, that is, because it is not economical to take such an action.

Risk

Plan – Positive Risks

Share. This can be used for both, negative risk (threat) or a positive risk (opportunity) type of risk. These responses will be included as part of creating the next stage plan or exception plan. This response is a form of risk sharing between two or more parties and is normally built into a contract. It uses some form of a pain/gain formula, and prescribed limits are used between the parties that divide up either the financial pain or gain if the opportunity or threat does not materialize.

Risk

Implement and Communicate

Implement. The risk responses identified above, are now implemented, and how effective each response is will be monitored and corrected where necessary to achieve the desired effect.

Communicate. Unlike the first four sequences above, this is a parallel and ongoing activity to ensure that information on all of the threats and opportunities are communicated both internally and externally to the project. The risks are generally communicated through time driven reports (such as Highlight Report, Checkpoint Report and so on) and event driven reports (such as end Stage report, end Project report and so on)

Risk

- The **Risk Owner** is responsible for managing & monitoring risks aspects. They can also carry out actions that have been assigned to them.
- The **Risk Actionee** is someone who is assigned to carry out a particular action and they support the Risk Owner. So they are not responsible for monitoring or managing the risk.
- The risk that remains after implementing a response to a particular risk is called as **Residual Risk** and the new risk that arise as a result of response to a particular risk is called as **Secondary Risks**

Risk

Risk Budget

Risk budget is a sum of money included within the project budget and set aside to fund specific management responses to the project's threats and opportunities (for example, to cover the costs of any fallback plans should they need to be implemented).

Residual and Secondary Risks

The risk that remains after implementing a response to a particular risk is called as **Residual Risk** and the new risk that arise as a result of response to a particular risk is called as **Secondary Risks**

Risk

Risk Tolerance and Risk Appetite

- Risk tolerance looks at acceptable/unacceptable deviations from what is expected.
- Risk appetite looks at how much risk a company is willing to accept. There can still be deviations that are within a risk appetite. A organization can be a Risk seeker, Risk Neutral or Risk Averse.
- Risk tolerance can be related to other tolerance parameters; risk to completion within time scale and/or cost and to achieving product quality and project scope within the boundaries of the Business Case.
- Perceptions of risk tolerance have to be considered in detail to establish the optimum balance of a risk occurring against the costs and value for money of limiting that risk.

Risk

- **Corporate or Programme Management** provides the corporate level policy for the risks
- **Executive is** responsible for ensuring that the risk management strategy exists and the risks associated with the Business case are identified, assessed and controlled.
- **Senior User is** responsible for ensuring that risks related to users are identified, assessed and controlled.
- **Senior Supplier is** responsible for ensuring that risks relating to supplier aspects are identified, assessed and controlled.

Risk

- **Project Manager** creates the Risk Management Strategy, creates and maintains the risk register. Ensures that the project risks are identified , assesses and controlled throughout the lifecycle of the project.
- **Team Manager** participates in the identification, assessment and control of risks at Work Package level.
- **Project Assurance** review risk management practices to ensure that they are performed in alignment with the project's Risk Management Strategy
- **Project Support** assists the project maintaining the project's risk register

Change

The purpose of the knowledge in the **Change Theme** is to help you identify, assess and control any potential changes to the products that have already been approved and baselined.

The Change Theme is not just about handling change requests but also handling issues that arise during the project. In fact, it is better to say that the Change Theme provides a common approach to issue and Change Control.

Change

Baseline

Refers to a product which has undergone its quality controls, has been deemed « fit for purpose » and has been approved by those with authority.

Information about their state is recorded in configuration item records.

Change

The PRINCE2 Manual uses the Change Theme to describe how change control should be executed. All changes are dealt with as a type of project issue. An issue can be

- General issues

- Request for Change

- Off Specifications.

Change

Request for Change

This is a change requested from the Customer/User side, and would, if implemented, cause a change to what had been originally agreed, to the Acceptance Criteria, Specifications/Scope.

It might be a request to add or subtract to the original agreement. If you were having a house built, two examples might be you requesting an extra bathroom, or asking for a dividing wall to be removed. As such, any extra costs relating to this change should be paid for by the Customer/Users.

Change

Off Specification

This covers errors or omissions either in work already carried out, or planned for the future. This will result in NOT being able to meet the originally agreed Acceptance Criteria, Specification/Scope.

An example similar to above would be if the builder of your new house advises that they can't include your patio area within the price. As such, any extra costs (either in re-work to fix the off- specification, or reducing the price to you), should be met by the builder. Suppose that (possibly in order to meet your timescale...) you agreed to accept what the builder could give you (that is, house without the patio), then in PRINCE2 terms, this is called a CONCESSION.

Change

General Issues

First of all, any general issue could be dealt with 'face to-face' if appropriate - logging it as a 'formal' project issue would be done if that were the best and only option. As an example, 'general' issues could include:

A question or query

A good idea or a suggestion

An observation

A concern

Change

Change Authority and Change Budget

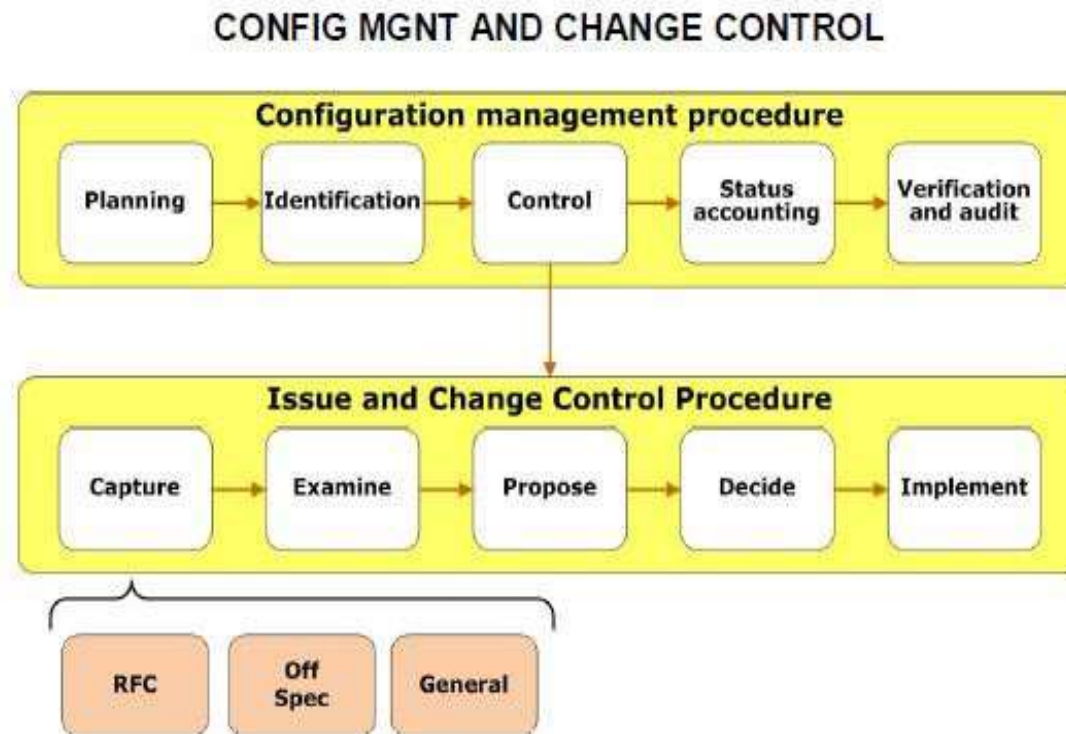
- The Change Authority is a person or a group who consider requests for change and off-specifications. It is the responsibility of the Project Board, so they can do it themselves, which is more common where few changes are expected, or they can assign this to other persons. If a lot of changes are expected then this will take up too much time from the Project Board and it is better to give the authority to another person or group of persons
- The Change Authority will have a change budget, which is a sum of money that the customer and supplier agree to use to fund the cost of Requests for Change. It is advisable to always have a change budget for each project unless you are sure there will be very few or no change requests. The Project Board can still exert control, as they can put a limit on the cost of a single change or the amount to be spent in any one stage.

Change

If an issue is forecast to exceed tolerances, it's an exception and requires escalation to the next highest level of management. Issues can be managed in 2 ways:

- 1) Formally – requires advice from project board – issues have to be logged into the issue register and an issue report is written
- 2) Informally – recorded in the daily log

Change Management Approach



Change Management Approach

Capture – either in the issue register or daily log

Examine – analyse the impact on cost, time, quality, scope , benefits, risks

Propose – alternative options for responding

Decide – which option is the best overall value for money

Implement – perform the recommended option

Capture

Issue register

To capture and maintain information on all the issues that are being formally managed. The issue register should be monitored by the project manager on a regular basis

Issue report

A report containing the description, impact assessment and recommendations for a request for change, off-specification or a problem concern. It is created only for those issues that need to be handled formally.

Change

Possible Options – Request for Change

Approve the Change

Reject the Change

Defer Decision

Request for more information

Ask for an Exception Plan

If a request for change requires an additional cost there are following ways to fund it

-Use Change Budget

-Increase the Project Budget

-De Scope other element of the Project Budget

Change

Possible Options – Off Specification Management

Grant a Concession

Instruct that Off- Specification must be resolved

Defer Decision

Request for more information

Ask for an Exception Plan

Change

Possible Options – Problems/ Concern Management

Provide Guidance

Ask for an Exception Plan

Change

There are three types of management product:

Baselines , records and reports .

Baseline management products are those that define aspects of the project and ,once approved are subject to change control. These are;

- Benefit Review Plan
- Business Case
- Communication Management Strategy
- Configuration Management Strategy
- Plan
- Product Description
- Project Brief
- Project Initiation Documentation
- Project Product Description
- Quality Management Strategy
- Risk Management Strategy
- Work Package

Change

Records are dynamic management Products that maintain information regarding project progress .

These are;

- Configuration Item Record
- Daily log
- Issue Register
- Lesson Log
- Quality Register
- Risk Register

Change

Reports are management Products providing a snapshot of the status of certain aspects of the project.

These are;

- Checkpoint Report
- End Project Report
- End Stage Report
- Exception Report
- Highlight Report
- Issue Report
- Lesson Report
- Product Status Account

Change – Role & Responsibilities

Corporate Programme Management provides the corporate level strategy for change control, issue resolution and configuration management

Executive determines the change authority and the change budget, sets the severity ratings and priority for issues, respond to request for advice from the project manager and make decision on escalated issues with the focus primarily on continual business justification.

Senior User takes decision on escalated issues with the primary focus on safeguarding the expected benefits.

Senior Supplier takes decision on escalated issue with primary focus on safeguarding the integrity of the complete solution

Change – Roles & Responsibilities

Project Manager manages the configuration management procedure, manages issue and change control procedure, creates and maintain the issue register

Team Manager implements corrective actions for Work Package level issues

Project Assurance advises on examining and resolving issues

Project Support maintains the configuration item records, produces product status account and assists the project manager to maintains the issue registers

Progress

- To establish how to monitor and then to compare actual achievements against those planned during the project life cycle.
- To provide a forecast for the project objectives and the project's continued viability.
- To be able to control any unacceptable deviations.
- Progress is about checking progress compared to the plan, checking project viability and controlling any deviations.

Progress

Three of the seven principles are represented in the Progress Theme; they are:

Manage by stages: the Project Board is to use stages as a control point.

Continued business justification, as the Business Case is continually checked that the project is still worth doing.

Managed by Exception. Where tolerances are used, refer certain issues up to the next management level.

Progress

Let us look at **when tolerances can be decided on:**

Time and Cost Tolerances: These are decided in the Project Plan, Stage Plans and Work Packages.

Scope Tolerances: Decided in Project Plan, Stage Plan and Work Packages.

Note: Scope changes would require change control.

Risk tolerances will be first defined in the Risk Management Strategy document and the Project Board can change risk tolerance for the Stage Plan. The Project Manager may change risk tolerances for the Work Package.

Quality Tolerances are defined in the Project Product Descriptions and the Product Descriptions, as Quality is related to the products.

Benefits tolerances are defined only in the Business Case and this is kept up to date during the project. The Benefits are also defined in the Business Case

Progress

The PRINCE2 approach to Progress and the four main controls provided by PRINCE2:

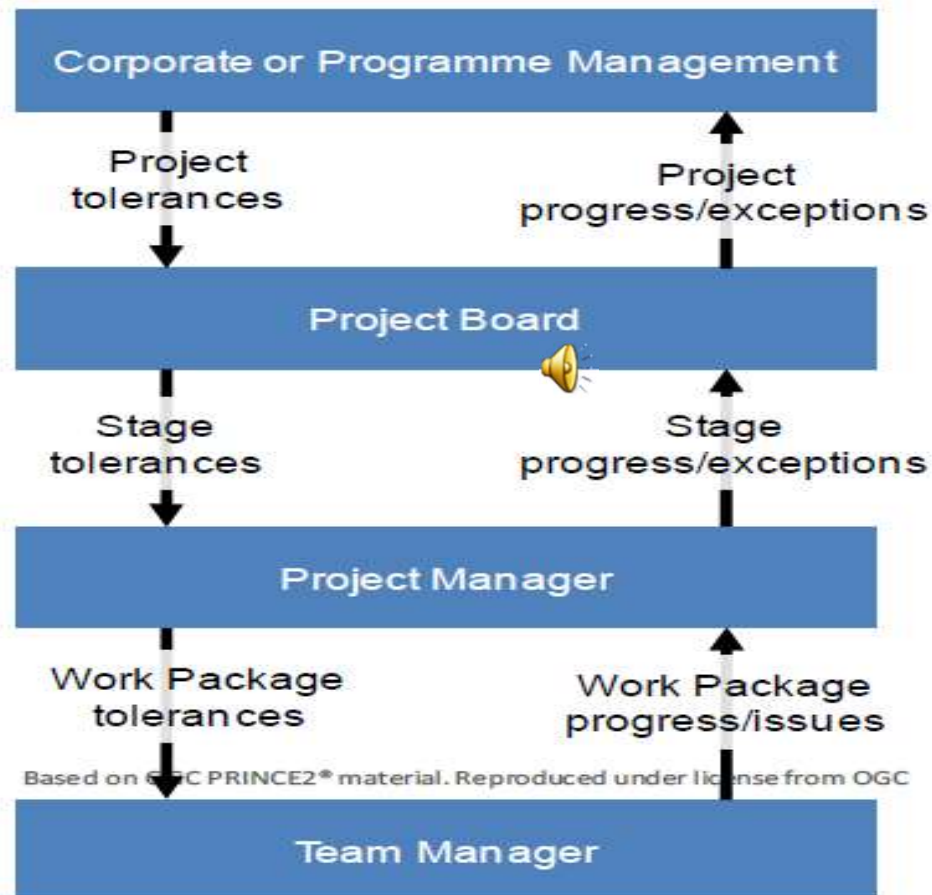
- (1) Delegating Authority,
- (2) Using Stages,
- (3) Time & Event-driven reports, and
- (4) Raising Exceptions Case

Progress

- All controls can be divided into two parts in PRINCE2: **Event-Driven and Time-Driven**.
- **Event-driven controls** take place when something happens, in other words when an event happens in the project. (For example, at the end of a stage, at completion of the PID, when a stage goes out of tolerance, at the end of project and change request. All of these events produce documents like an End Stage Report, Exception Report and Issue Report.
- **Time-driven controls** take place at pre-defined periodic intervals. For example, the Project Board will agree with the Project Manager to send a **Highlight Report** every 2 weeks to the Project Board, and the Project Manager can agree with the Team Manager to send a **Checkpoint Report** each week. So time-driven controls don't have to wait for an event to happen.

Progress

Delegating Tolerance and reporting process



Progress

Why are Management Stages used as controls by the Project Board?

- Management stages are partitions of the project with decisions points for the Project Board between each stage. A management stage is a collection of activities to produce products and is managed by the Project Manager.

Why are Management Stages important for the Project Board?

- They provide review and decision points at end of each stage and before the next stage. They can authorize one stage at a time, or choose to stop the project.
- They review the End Stage Report of the last stage and Review plan for next stage.
- Then can check project progress compared with baselined Project Plan at the end of each stage.

Progress

The minimum number of stages in a PRINCE2 project is two: the Initiation Stage to define and agree what needs to be done, and at least one other stage to produce the products.

Progress

How to decide the number of stages?

- This depends on a number of items and as you can see, it's a bit of a balancing act. Start by considering the following
- How far ahead is it sensible to plan?
- Where do key decision points have to be made in the project? (Example: Maybe after creating a prototype or after completion of a major part of the product. This would be a good point for stage end.)
- The amount of risk and the complexity in a project. (If similar to another project, then there will be less.). Higher the risk and complexity more will be the number of management stages.
- The number of management stages in a project, and is a bit of balancing act as the more number of management stages provide better control on the project, however increases the administrative overhead.

Progress – Roles & Responsibilities

Roles and Responsibilities

- **Corporate programme management** provides the project tolerances and the document them in project mandate and make decisions related to exceptions when project – level tolerance is forecast to exceed.
- **Executive** provides stage level tolerances and makes decision when stage level tolerances are forecast to be exceeded. Ensures that progress towards the outcome remain consistent from business perspective. Recommend future action on the project to corporate or programme management if the project tolerance is forecast to be exceeded
- **Senior User** ensures that progress towards the outcome remain consistent from user perspective.
- **Senior Supplier** ensures that progress toward the outcome is consistent from the supplier perspective

Progress – Roles & Responsibilities

Roles and Responsibilities

- **Project Manager** authorizes the Work Package level tolerances and monitor progress against the stage plan. Produces highlight reports, End Stage report, lesson report and end project report. Produce exception reports when the stage level tolerances are forecast to be exceeded.
- **Team Manager** agrees on work package with the project manager, produce checkpoint reports and notify the project manager of any forecast deviation from work package tolerance.
- **Project Assurance** review and verify the business case against the external events, verify impact on the business case on the basis of progress or due to change in the plan.
- **Project Support** assist with the compilation and distribution of reports, assist the project manager in maintaining the issue and risk registers.